

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		transfer of electrons – one calcium atom loses two electrons AND one oxygen atom gains two electrons (1) ions – one Ca ²⁺ ion AND one O ²⁻ ion with eight electrons around it (1) if inner shells drawn all atoms and ions must be correct		2		2		
		(ii)		any of following for (1) • strong bonds between ions • strong ionic bonds • strong electrostatic forces between ions neutral answer ‘strong bonds’ either of following for (1) • attraction between ions with greater charge is greater • 2+/2– attraction is greater than +/- attraction	2			2		

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					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		(each carbon atom) only bonded to 3 other carbon atoms (1) do not award first mark if any reference to metallic bonding delocalised electrons able to move (through structure) (1)	2			2		
		(ii)		9.1×10^{-10} (3) accept 0.91×10^{-9} if incorrect award (1) for each of following $11 \times 0.26 = 2.86$ diameter = circumference $\div \pi$ / $\frac{2.86}{3.14}$ ecf possible		3		3	3	
				Question 4 total	4	5	0	9	3	0